SERVER:
OPEN CHIPLET
ECONOMY

Advancing Chiplet Flexibility: Updates in 2.0 OCE Transaction and Link Layer Specification

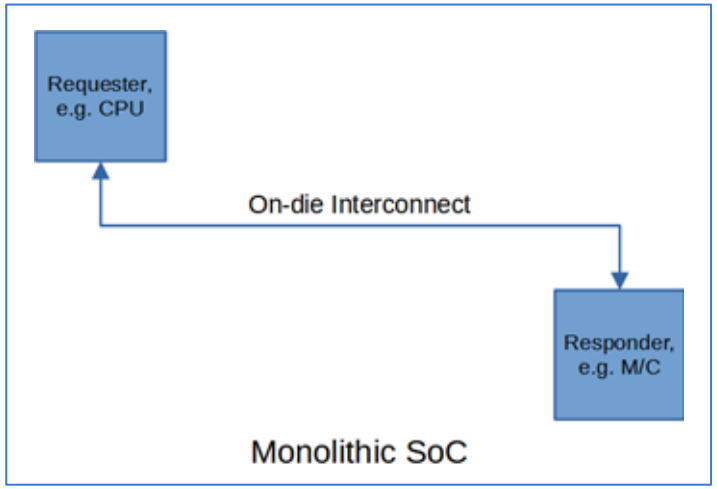
Brian Murray

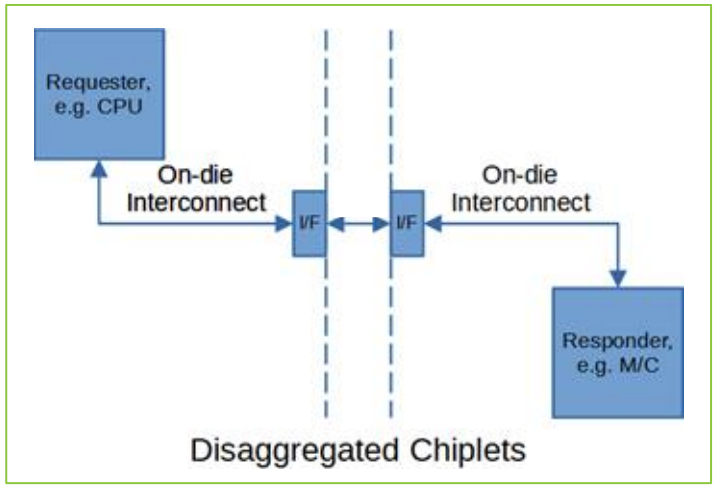
SoC Architect, VeriSilicon

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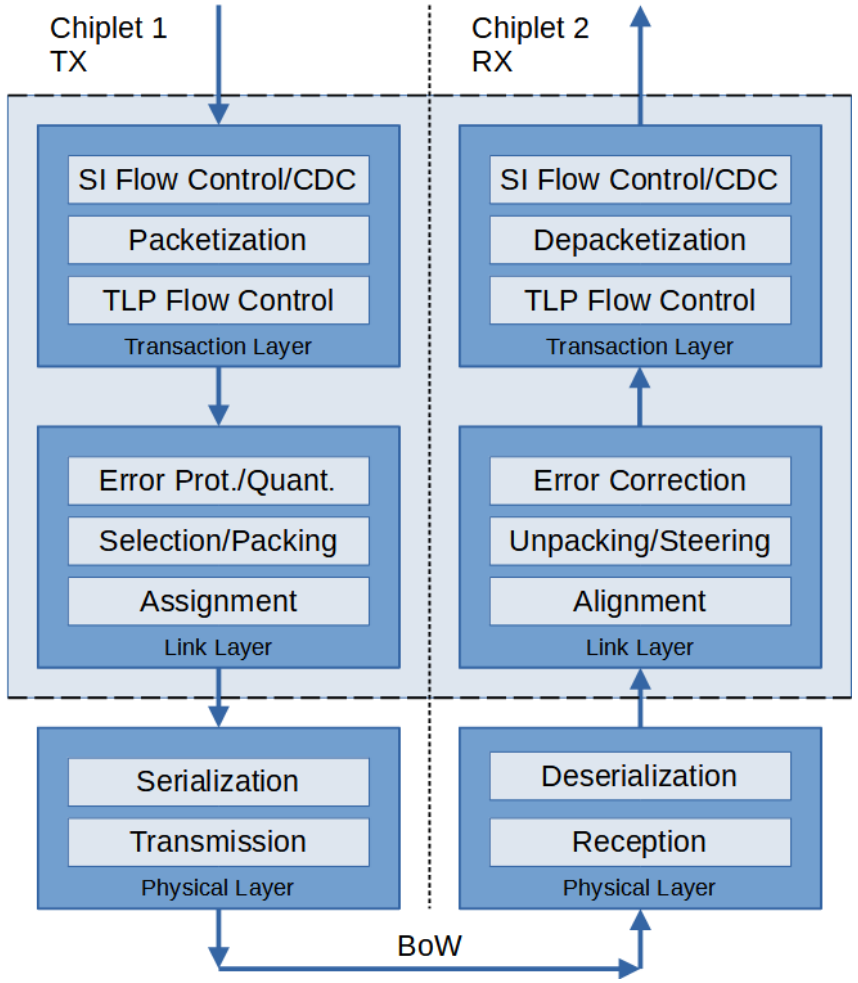
OCE Transaction and Link Layer

- Chiplet Die Disaggregation
  - Any IP Protocols become chiplet interfaces
  - Agnostic to chiplet architecture
  - Transparent to software
- Independent of Transport
  - Supports mapping to a wide variety of physical layers
- Chiplet SiP = Monolithic SoC
  - Once chiplet links are brought-up, the chiplet integration behaves like a monolithic SoC





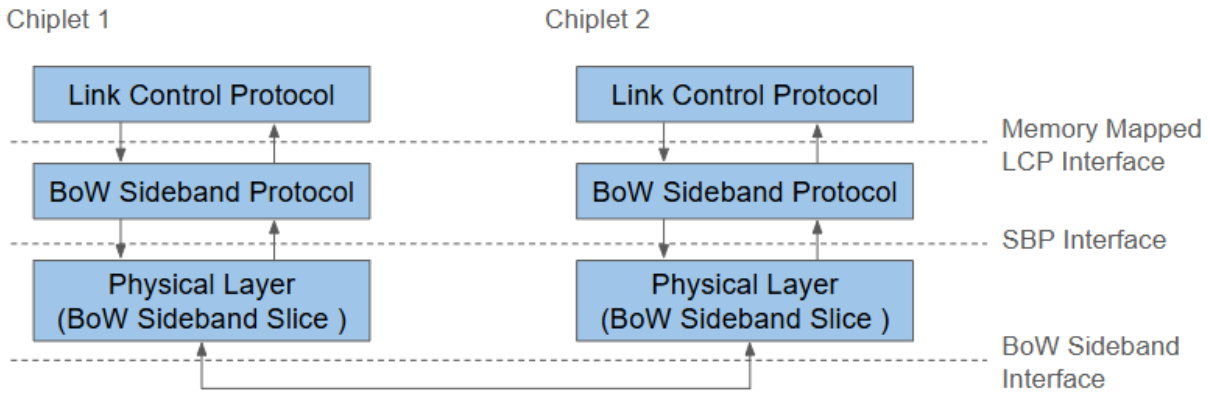
Background: 1.0 Transaction and Link Layer



- A protocol stack for translating bus transactions across a BoW interface between two chiplets
  - Transport for a variety of busses and signals
  - Focus on simplicity, flexibility, and low latency
- Transaction Layer maps SoC bus protocols (e.g. AMBA) to message packets
  - Wide support that is easily extensible
- Link Layer adds error protection (ECC) and breaks the messages into granules
  - Maps varying width BoW “Mainband” interfaces

Background: 1.2 OCE Transaction and Link Layer

- 1.2 Introduces a protocol stack that utilizes the BoW Sideband Slice that complements the Mainband Link Laye
  - Addition of a BoW Sideband Protocol
  - Addition of a Link Control Protocol



- General fixes and clarifications in the Mainband link layer specification
  - No functional changes to the existing text
  - No issues with compatibility with the 1.0 spec

2.0 OCE Transaction and Link Layer Summary

- Revisit all TLL spec decisions
  - Reduce all overheads
    - Fewer reserved and AUX bits
  - Reconsider all limitations
    - E.G. TLP rate limits
  - Limited compatibility with 1.x
    - Only Link Layer has compatibility modes
- Revised Packet Formats
  - TLP headers
  - All Uncredited TLP Classes
  - LLP header and LLP format
- Support for PHY Binding
  - Generalized Link Layer to better accommodate a wider range of physical layers
  - Adding **UCIe** and **BoW.FLEXI** support
- Better guidance for Variants and Profiles
  - More clarity and flexibility with options in how adjunct specifications can affect implementations

Revisiting the Transaction Layer Packet Format

- General Streamlining
  - The TLP header can be transmitted very frequently, so efficiency is paramount
- 1.x TLP Format
  - 12b Format including 6b Type, 1 Reserved, and 5 ‘AUX’ bits
  - Extra bits could be used by Profiles, but usage was inconsistent
- 2.x TLP Format
  - 4b Type and 2b Type Extension
  - Profiles can choose whether the Type Extension is used to extend the Type value or as Payload

New View of Uncredited TL Packets

- All Uncredited Types
  - Now rate limited by the Profile
    - Not limited to 1 TLP per LLP
  - Smaller TLP header = increased payload %
- MSG: Messages
  - Message payloads up to 16 bytes, with byte granularity
  - Ability to target different message destinations
- VWX: Virtual Wires
  - No change
- CRD: Credit Returns + IDLE
  - 2.x CRD has three Modes
    - 1.x like format indexed by TLP Type, but now with two credit returns per TLP
    - Multi-credit fields (5) defined by Variant/Profile, indexed by a 4b Bank ID
      - Meant to replace the 1.x AUX crediting scheme
    - IDLE (CRD with no credits)
      - Replaces the 1.x IDLE Type
  - Change to a 3b credit field encoding
    - Profile defined encoding
    - Default Int/2^(n-3) encoding provides best trade-off between number of credits and number of fields

Link Layer Changes

- Specification Reformatting
  - Cleaner delineation between the Transaction and Link Layers
  - Easier to read the UCIe and BoW.FLEXI Link Layers without referencing the BoW sections
  - Link Layer Generalization – More flexible concepts for describing bindings, such as PHYs with frame signaling
- Link Layer Packet formats
  - 24 Granule sized LLP (1 Header and 23 Payload) – 2.x profile default
    - Relaxed/removed TLP packing constraints within an LLP
    - LLP size increase reduces LlpHdr transmission overhead improving the link efficiency
  - 16 Granule sized LLP (1 Header and 15 Payload) option
    - Supported for limited 1.x backwards compatibility and systems with lower latency requirements

New Link Layer Mappings

- UCIe RAW Streaming
  - Utilizes the “Universal TLL” proposal from Ventana
  - Maps the existing TLP format to the UCIe FDI
    - Includes granule ordering that respects both TL and UCIe fundamentals
- BoW.FLEXI
  - Supports the frame signal defined packetization from BoW.FLEXI
  - Layers on top of the existing FLEXI spec by utilizing the EXT0 command
    - The FLEXI header replaces the LLP header
    - Utilizes the FLEXI ECC error correction
  - No ‘start’ bits
    - Due to the frame signal, the TLP to LLP ratio is typically 1:1 for lower width interfaces

Adjunct Documentation Guidance

- Bus Protocol “Variants”
  - Additional clarity on credit groups
  - Allow the Variant authors to better guide its use in Profiles
- Interface “Profiles”
  - Includes more clarity on values that need to be defined to guide hardware implementations
  - Flexibility on Type size
    - The Type Extension can optionally be used as Payload by the Profile
  - Flexibility to exceed baseline rate limits
    - E.G. a Profile with many Types can allow more CRD packets per LLP
    - Allows systems to tradeoff implementation costs, Stream QoS flexibility, and LLP packing efficiency
  - More flexibility on credit returns and reuse of guidance from Variants
  - Flexibility on LLP size

Come Join Us!

- Road to Chiplet Interoperability
  - **Join** Open Chiplet Community!
  - **Attend** OCP/OCE Link Layer and System weekly calls
  - **Contribute** to Variants and Profiles specifications and the next TLL version
  - **Build** an interoperable OCP Chiplet Marketplace

- Where to find additional information (URL links)
  - Project Wiki: http://www.opencompute.org/wiki/Server/OpenChipletEconomy/Link-Layer
  - Marketplace: https://www.opencompute.org/chiplets

Thank You!

